



IoT-Based Smart Infusion Monitoring and Control System Using ESP32

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Abstract

Infusion is a common medical procedure used to treat conditions such as gastric acid and typhoid, where precise fluid administration is critical. This study presents the development of an IoT-based smart infusion monitoring and control system using an ESP32 microcontroller, designed to automatically monitor infusion volume and regulate drip rate in real-time. The system integrates a load cell sensor to measure infusion fluid weight, a photodiode sensor to detect drip rate, and a servo motor to adjust the flow rate adaptively. It features web-based monitoring, buzzer alerts, and an LCD display for local feedback. The system was tested in a clinical simulation with an infusion requirement of 1500 mL per 24 hours and various drip factors (15, 20, and 60 drops/mL). The infusion volume status is automatically categorized into three levels: FULL (>350 mL), HALF (150–350 mL), and WARNING (<150 mL). Based on 10 test scenarios, the system accurately classified volume levels and triggered warnings when volume dropped below 150 mL. For example, in Test-08 to Test-10, volumes of 139.67 mL, 87.34 mL, and 40.53 mL were correctly detected as “WARNING” with buzzer alerts activated. The load cell sensor achieved excellent accuracy, with an error margin between 0.02% and 0.06%, while the system maintained drip-rate stability within a $\pm 5\%$ tolerance range. It also dynamically adjusted the servo angle to correct under- or over-drip conditions. These results confirm that the system delivers accurate, automated, and responsive infusion control, making it suitable for healthcare settings with limited staff to improve safety and efficiency.

Keywords: automatic control; ESP32; infusion fluids; internet of things; monitoring system

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1. Introduction

The infusion system is a crucial component in patient care, particularly in the intravenous (IV) administration of fluids, drugs, and nutrition [1], [2]. In many hospitals in Indonesia, including RSU Porsea and RS HKBP Balige in the Toba region, infusion management is still performed manually [3]. Manual systems are prone to human errors such as inaccurate drip-rate settings, delayed fluid replacement, and a heavy reliance on continuous visual observation. These limitations increase the risk of treatment delays, particularly in facilities with limited medical personnel, where excessive workloads are common [4], [5]. Patient movement further complicates infusion therapy, as it can alter device positioning and disrupt fluid flow, requiring manual adjustments that may not always be performed promptly [6]. Consequently, there is an urgent need for an automated and real-time infusion

monitoring and control system to improve efficiency and patient safety [7] - [10].

Several studies have attempted to address this challenge using microcontroller-based systems and Internet of Things (IoT) technologies. Rosyady et al. [11] developed an Arduino Uno-based system using a load cell and real-time clock (RTC) module to estimate infusion expiration time, with outputs displayed on an LCD and local buzzer notifications. However, it lacked IoT connectivity and automatic flow control. Firdaus et al. [12] designed a system using an ATmega328 with TCRT5000 sensors and local wireless communication, but it could not adaptively adjust infusion rates and did not support remote notifications. Caya et al. [13] combined Raspberry Pi, Arduino, and optical sensors with a Tkinter-based desktop GUI, which enabled infusion control and data storage in a MySQL database. Despite these features, the system supported only local

monitoring and provided limited alerts when the remaining fluid volume dropped below 10%.

Other researchers have also explored alternative approaches. Wijayanto et al. [14] implemented an Arduino Mega2560 and load cell-based monitoring system with server connectivity, but it lacked automated flow control and integrated alarms. Firmansyah et al. [15] developed a photodiode- and servo-based monitoring system with MQTT communication, but it relied solely on drop detection without considering fluid volume and did not provide comprehensive web monitoring or adaptive adjustment. Maharani et al. [16] proposed a web-based monitoring and warning system using Arduino Uno, load cell, LED, and photodiode sensors, achieving a low error rate but offering limited alerting functionality. Another study employed LabVIEW and fuzzy logic with ATmega controllers and stepper motors, achieving precise control but relying on expensive and less accessible hardware platforms.

From these studies, it is evident that most existing systems focus only on either monitoring or control, rely heavily on local interfaces (LCDs or PCs), lack adaptive real-time control, and do not integrate IoT-based remote access. These limitations restrict their applicability in clinical environments where staff resources are limited and timely responses are critical.

To overcome these gaps, this study proposes the development of an ESP32-based intelligent infusion monitoring and control system that integrates both monitoring and regulation in real time. The ESP32 platform is selected for its dual-core processor (up to 240 MHz), built-in Wi-Fi and Bluetooth, energy-efficient operation modes, larger memory capacity, and higher number of GPIOs compared to alternatives such as Arduino Uno or ESP8266 [17] - [21]. Additionally, ESP32 supports multiple development environments, including Arduino IDE, PlatformIO, and MicroPython, offering flexibility for IoT applications.

The objectives of this study are to: (1) enable real-time monitoring of infusion fluid volume using a load cell sensor; (2) provide automatic drip-rate regulation via a servo motor actuator; (3) integrate a web-based monitoring platform accessible remotely by medical personnel; and (4) implement local and online notifications for critical conditions such as low fluid levels or mismatched flow rates [22], [23]. A case study involving patients with stomach acid and typhoid conditions is used to evaluate the system's performance, demonstrating its potential to improve accuracy, efficiency, and responsiveness while reducing the workload of medical staff.

The novelty of this research lies in the integration of monitoring and control functionalities into a single IoT-based system designed for resource-limited healthcare environments. Unlike previous works, the proposed system combines adaptive servo-driven drip-rate adjustment, IoT-based real-time monitoring and

notification through MQTT, classification of infusion volume status (FULL, HALF, WARNING), and hardware optimization using the ESP32 microcontroller. Experimental results show high accuracy in volume detection (0.02–0.06% error) and stable drip-rate control within $\pm 5\%$ tolerance, outperforming comparable IoT-based systems [16]. This contribution provides a more reliable, accessible, and scalable solution for smart infusion management in clinical settings.

2. Methods

This study employs a Research and Development (R&D) approach to design, implement, and evaluate an Internet of Things (IoT)-based intelligent infusion monitoring and control system. The methodology consists of three main stages: system design, prototype development, and testing.

2.1 Standard Infusion Volume Status

Continuous monitoring of infusion fluid volume is very important in the management of infusion therapy, to ensure that patients receive timely therapy and according to medical needs. A decrease in infusion fluid volume that is not properly detected can lead to inadequate fluid administration, even the risk of running out of infusion fluid which can endanger the patient's condition. Therefore, an automatic system that can detect and classify infusion fluid volume status is very necessary to improve patient safety.

In this study, infusion volume status is categorized into three main statuses: Full, Half, and Warning, based on the volume of infusion fluid remaining in the infusion bottle. When the fluid volume reaches Warning status, the system will provide a notification via buzzer and Serial Monitor to notify medical personnel. The following are the infusion volume standards used in the system, based on an analysis of applicable hospital guidelines and nursing practices: Full infusion volume, Full status is set when the infusion fluid volume is above 70% of the total capacity of the infusion bottle. In this status, the infusion fluid is considered quite full, and no special action is required other than routine monitoring. For example, for a 500 ml infusion bottle, the Full status will be applied if the infusion fluid volume is above 350 ml (70% of 500 ml). Half infusion volume, The Half status is set when the infusion fluid volume is between 30% and 70% of the total capacity of the infusion bottle. At this status, the infusion fluid has started to decrease, and more frequent monitoring is required. This is an early signal for medical personnel to prepare to replace the infusion bottle. For a 500 ml infusion bottle, the Half status occurs when the infusion fluid volume is between 150 ml and 350 ml (30% and 70% of 500 ml). Warning infusion volume, The Warning status is set when the infusion fluid volume is below 30% of the total capacity of the infusion bottle. At this stage, the fluid volume is approaching the critical limit, and immediate action is required. The system will activate the buzzer and display a warning

on the Serial Monitor to notify medical personnel that the infusion bottle must be replaced immediately. For a 500 ml infusion bottle, the Warning status occurs when the infusion fluid volume is below 150 ml (30% of 500 ml).

Based on the results of this analysis, the standard infusion volume status used in the implementation and testing is shown in Table 1.

Table 1. Standard Infusion Volume Status

No	Infusion Volume (500 ml NaCl)	Status	Notification
1	>350 ml	Volume FULL	LCD: FULL Buzzer: OFF
2	150 ml – 350 ml	Volume HALF	LCD: HALF Buzzer: OFF
3	<150 ml	Volume WARNING	LCD: Change IV Fluid Buzzer: ON

As shown in Table 1, the standard provisions for infusion volume status are defined to enable timely notifications, allowing medical personnel to take quick and appropriate action in replacing the infusion bottle or adjusting the infusion flow.

2.2 Drop Factor Analysis and Number of Drops per Minute

In the infusion therapy management system, one of the crucial aspects that must be considered is the drop factor and the calculation of the number of drops per minute (DPM). A proper understanding of these two parameters is essential to ensure that patients receive fluid volume accurately and in accordance with medical prescriptions. Inconsistencies in fluid administration can lead to the risk of overdose or underdose, which can potentially endanger the patient's condition [7]. The drop factor is a measure that indicates the number of drops needed to produce one milliliter (mL) of fluid from the infusion. This value depends on the type of infusion device used and is usually expressed in drops per milliliter (dpm/mL).

Meanwhile, the number of drops per minute (DPM) is the flow rate of the infusion fluid calculated based on the volume of fluid per hour needed by the patient and the drop factor of the infusion set.

Type of Drop Factor: Micro Chip has a drop factor of 60 drops/mL. Generally used for pediatric patients or administration of drugs with small doses that require a high level of precision. Otsuka Brand Macro Chip: Has a drop factor of 15 drops/mL. Generally used in adult patients who require large volumes of fluids or in conditions that require rapid resuscitation. Terumo Brand Macro Chip has a drop factor of 20 drops/mL, and is also used for adult patients with standard infusion needs. Infusion Needs for Acid Reflux and Typhus Patients. In adult patients suffering from acid reflux and typhus, the daily infusion fluid requirement is 1500 mL per 24 hours. Commonly used infusion fluids are 0.9% NaCl or Ringer Lactate, with a pattern of administration of 1 bottle containing 500 mL every 8 hours.

$$\text{Drip Rate (dpm)} = \frac{\text{Volume of IV fluid (mL)}}{\text{Time to Run (h)}} * \frac{\text{Drop Factor} \left(\frac{\text{drops}}{\text{mL}} \right)}{60} \quad (1)$$

The drip rate of an infusion can be calculated using Formula 1. It is determined by dividing the total volume of IV fluid (in milliliters) by the prescribed infusion time (in hours), and then multiplying the result by the drop factor of the infusion set (in drops per milliliter). Finally, the value is normalized by dividing by 60 to convert hours into minutes. In this context, the volume of IV fluid represents the total liquid to be administered, time to run is the duration over which the fluid must be delivered, and the drop factor depends on the type of infusion set being used. This calculation ensures that the infusion drip rate (in drops per minute) aligns with the prescribed medical dosage.

As shown in Table 2, the number of drops per minute for acid reflux and typhus diseases was calculated based on Formula 1.

Table 2. DPM of Gastric Acid Disease and Typhoid Based on Type of Drip Factor

No	Type of Drip Factor	Drip Factor (drops per ml)	Drops Per Minute (DPM)
1	Micro Chip	60	62.5
2	Macro Chip Otsuka	15	15.6
3	Macro Chip Terumo	20	20.8

With this analysis, the developed system can be adjusted to the type of infusion set used and the patient's fluid needs, so that monitoring and control of the infusion rate can be carried out more precisely and adaptively.

2.3. Analysis of Drip Factor and Drops per Minute

In designing an automated system to regulate the infusion fluid drip-rate, determining the tolerance limit is an important factor that affects the performance of the system to remain safe, effective, and in accordance with medical needs. This tolerance functions as a deviation limit that is still considered reasonable from the predetermined drip-rate target. If the system is too sensitive to small fluctuations, this can trigger unnecessary alarms, while if the tolerance is too wide, the risk of dose errors will increase.

Several considerations that affect the determination of tolerance include, Type of Infusion Fluid: Fluids with higher viscosity, such as total parenteral nutrition solutions, usually require looser tolerance limits than regular saline solutions. Patient Condition: For patients in critical condition, the infusion rate must be controlled with high precision. Therefore, the tolerance is set tighter, for example around $\pm 2\%$ of the drip-rate target. Type of Clinical Therapy: In therapies that are very sensitive to dose, such as chemotherapy or administration of heart drugs, much more precise drip control is required than regular infusions for hydration.

Formula for Determining Tolerance Limits: To calculate the upper and lower limits of tolerance for the desired drip-rate, a specific formula is used that is designed to keep the infusion within a safe and effective range.

$$R_{\text{upper}} = R + (T * R) \dots \text{Upper Limit Tolerance} \quad (2)$$

$$R_{\text{lower}} = R - (T * R) \dots \text{Lower Limit Tolerance} \quad (3)$$

$$R_{\text{range}} = R_{\text{lower}} \text{ upto } R_{\text{upper}} \dots \text{Range Tolerance} \quad (4)$$

In Formulas 2, 3 and 4, R denotes the desired drop rate in drops per minute, while T represents the allowable tolerance, expressed as a percentage (e.g., a tolerance of 5% corresponds to $T=0.05$). For example, if the expected drop rate is set to 20 drops per minute with a tolerance of 5%, the system defines an upper limit (R_{upper}) and a lower limit (R_{lower}) for acceptable variation. The upper limit is calculated as $R_{\text{upper}}=20+(0.05 \times 20)=21$ drops per minute, while the lower limit is determined as $R_{\text{lower}}=20-(0.05 \times 20)=19$ drops per minute. This ensures that the infusion flow remains within an acceptable range of 19–21 drops per minute, providing both safety and flexibility in clinical application.

By setting this tolerance, the system will still consider the drop rate to be within reasonable limits as long as it is within that range and will only activate automatic correction or alarm if there is a deviation outside that limit. Applying this tolerance allows the system to: Reduce the number of unnecessary corrections, increase operational efficiency, Ensure patient safety and comfort.

2.4 Servo Motor Calibration with Infusion Fluid Drip Rate

One of the crucial aspects in the implementation of an automatic infusion control system is the servo motor calibration process, which functions to adjust the valve opening angle so that it can achieve the right infusion drip-rate. Calibration is carried out to ensure that the system is able to adapt to variations in physical conditions, such as fluid pressure, viscosity, or the height of the infusion bag position.

The servo motor calibration procedure involves preparing the system to ensure that all components, such as the servo motor, photodiode and LED sensors, control valves, and controllers, are properly installed and functioning correctly. Fill the infusion bag that will be used as calibration fluid. Make sure the infusion hose is properly connected. The system initialization process is Load Cell Infusion Tube Photodiode & LED Sensor Servo Motor Infusion Pole Brain LCD System Infusion Set Monitoring Web Application Set the initial servo angle, for example 45 degrees. Calibration Process to run the system so that the fluid will start to drip, the Photodiode and LED sensors begin to detect each drop that passes through the infusion line. The controller will count the number of drops that occur in one minute. Servo angle measurement and adjustment to record the

number of drops per minute produced at the initial servo angle (45 degrees). Example: Initial servo angle 45 degrees, the number of drops is 40 drops per minute. Compare the results obtained with the target number of drops, for example 50 drops per minute. If the number of drops is less than the target, increase the servo angle to open the valve wider (for example, from 45 degrees to 50 degrees). If the number of drops is more than the target, reduce the servo angle to close the valve slightly (for example, from 45 degrees to 40 degrees). The loop measurement process is carried out After adjusting the servo angle, repeat the measurement of the number of drops per minute. Make small adjustments to the servo angle until the number of drops produced approaches the target number of drops per minute in Table 2. The process of determining the optimal angle is carried out After reaching the drop rate that matches the target in Table 2, record the servo angle used. This servo angle will be a reference for the next system initial setting. Verification and Stability, Run the system for several time cycles (e.g. 5-10 minutes) in order to ensure that the number of drops per minute remains stable at the desired number using the same servo angle. Analyze if there is any variation in the number of drops and make adjustments if necessary.

The results of the servo motor calibration for the infusion drip-rate, obtained from the calibration process, are shown in Table 3.

Table 3. Servo Motor Calibration for Infusion Drip-Rate

Type of Drip Factor	Infusion Requirement	Drip-Rate (drops per minute)	Servo Angle
Micro Chip	500 ml / 8 jam	62.5	87 ⁰
Macro Chip	500 ml / 8 jam	15.6	45 ⁰
Otsuka			
Macro Chip	500 ml / 8 jam	20.8	56 ⁰
Terumo			
	80 ml / 1 jam	26.6	59 ⁰
	125 ml / 1 jam	41.6	64 ⁰
	126 ml / 1 jam	42	66 ⁰
	167 ml / 1 jam	55.6	75 ⁰
	2400 ml / 24 jam	33.3	61 ⁰
	3200 ml / 24 jam	44.4	68 ⁰

Specifically, our system employs a closed-loop feedback control approach, which is a fundamental principle in classical control theory. The photodiode sensor continuously measures the real-time drip-rate, and the microcontroller (ESP32) compares this value to a predefined setpoint (based on patient needs and clinical standards). If there is a deviation beyond the tolerance threshold, the system dynamically adjusts the servo motor angle to correct the flow. This behavior follows the feedback-control model where the system output (drip-rate) is monitored and used to regulate the actuator (servo motor) in real time.

2.5 System Design

At this stage, hardware and software architecture design is carried out. Hardware includes integration between the Load Cell to measure the volume of infusion fluid, Photodiode and LED sensors to detect drops, and servo

motors as infusion rate control actuators. Software involves programming on ESP32 and developing a web-based monitoring interface. The system architecture is designed to enable real-time monitoring and automatic notification delivery when critical conditions occur.

We selected the ESP32 microcontroller due to its robust processing capability and integrated wireless communication features, which are essential for real-time IoT-based medical monitoring systems. The specifications of the ESP32 that support our system design include: Processor: Dual-core Tensilica LX6 CPU, up to 240 MHz clock speed. Memory: 520 KB SRAM and external flash support. Connectivity: Built-in Wi-Fi (802.11 b/g/n) and Bluetooth v4.2 for seamless data transmission. GPIOs: Up to 34 programmable input/output pins, enabling integration with multiple sensors and actuators. Power Management: Equipped with energy-saving features like deep sleep and light sleep modes. Development Tools: Supports Arduino IDE.

The architecture of the ESP32-based IoT system for automatic infusion monitoring and control is represented by the block diagram in Figure 1, which consists of four main layers: Device Layer, Communication Layer, Application Layer, and Data Layer.

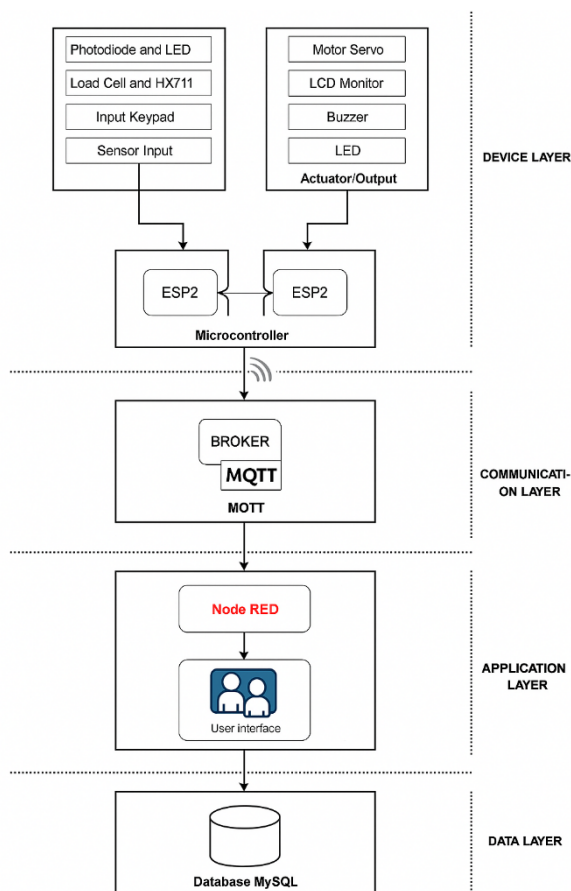


Figure 1. Proposed System Block Diagram

In the Device Layer, there are two ESP32 modules, which serve as the brain of the system for the microcontroller center (Microcontroller), connected to various input and output components. Input components in the form of Photodiode and LED are used to detect infusion drops, Load Cell and HX711 to measure the weight or volume of infusion fluid, and Keypad as manual input for system settings. Meanwhile, the output components or actuators include a Servo Motor to regulate the infusion valve, an LCD Monitor to display information, a Buzzer as a warning alarm, and an LED as a visual indicator. Furthermore, data from the microcontroller is sent wirelessly using the MQTT protocol located in the Communication Layer. The automatic infusion monitoring and control system developed in this study employs the MQTT (Message Queuing Telemetry Transport) communication protocol as the core of its data communication architecture. MQTT was selected because it is a lightweight, efficient publish/subscribe protocol, highly suitable for Internet of Things (IoT) applications that require real-time data communication with limited bandwidth. In this system, data communication occurs between the ESP32 microcontroller, located in the device layer, and the Node-RED-based application platform, located in the application layer. The mechanism begins with the ESP32 functioning as a publisher, sending sensor data such as the number of infusion drops detected by the photodiode and LED, as well as infusion volume data measured by a Load Cell and HX711 module. This data is transmitted as payloads to specific topics through an MQTT broker operating in the communication layer. The MQTT broker in this system acts as an intermediary, delivering messages between the publisher (ESP32) and the subscriber (Node-RED). Each sensor data message published by the ESP32 is received by the broker and subsequently forwarded to the subscribers registered to the corresponding topics. In this case, Node-RED acts as the subscriber, receiving data from the MQTT broker and processing it for visualization in the user interface and storage in a database. MQTT offers several advantages that support the system's stability and efficiency, including: Lightweight and low bandwidth usage, ideal for microcontrollers with limited resources. A publish/subscribe mechanism that supports one-to-many communication. Configurable Quality of Service (QoS) levels to ensure reliable data transmission.

Overall, the use of MQTT in this system enables real-time data exchange between the device and application layers. This capability is critical for the automatic and remote monitoring and control of infusion systems, such as issuing alerts through a buzzer and displaying notifications on an LCD screen when the infusion volume is nearly depleted or when a disruption in the drip flow occurs. Through this architectural approach, the system efficiently integrates various sensors and actuators with a web-based monitoring process, establishing MQTT as a reliable communication

backbone in IoT applications for healthcare. The connected User Interface allows users to monitor and control the system via desktop or mobile devices in real-time. Finally, all measurement data and system status are stored in a structured manner in the MySQL Database in the Data Layer, so that further recording and analysis can be carried out. The integration of these four layers forms an intelligent automatic infusion monitoring and control system that can be accessed remotely.

Test Conditions, to validate the system functionality and performance, the following test procedures and conditions were applied: Load Cell Sensor Accuracy, Infusion Volume Detection, Drip-Rate Monitoring and Automatic Control, Notification System by LCD, Buzzer and Web.

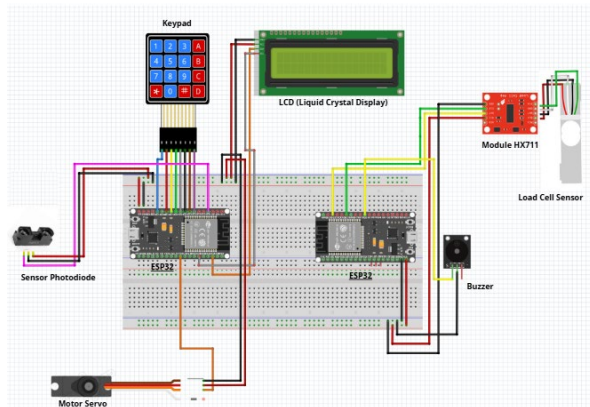


Figure 2. Hardware Design: Circuit Diagram

Figure 2 presents the circuit diagram of the system implemented in this study, which includes two ESP32 microcontrollers that serve as the central controllers of the system. The first ESP32 is connected to the photodiode sensor and keypad, servo motor actuator, and LCD. The second ESP32 is connected to the load cell sensor and HX711 module and buzzer actuator. These two ESP32s will communicate with each other via UART protocol communication. This circuit design will be implemented on a printed circuit board (PCB) to reduce the risk of connection errors, improve space efficiency, and achieve a neat component arrangement, simplify system integration, and enhance aesthetics and professionalism.

Figure 3 illustrates the physical design of the automatic infusion device integrated with a microcontroller-based monitoring and control system. This design resembles a standard IV pole modified by adding several main system components. The IV Pole is 200 cm or 2 meters high and 2 cm in diameter, made of sturdy material to support various system components and infusion devices. At the top there is a horizontal pole 40 cm long and 2 cm wide used to place the Load Cell sensor so that it can measure the volume of infusion that will be hung on the top pole. Right below the infusion bag, a 10 cm x 4 cm Infusion Drip Regulator is installed which functions to control the flow of infusion drops. This regulator is directly connected to the servo motor,

which will regulate the flow based on signals from the control system. A drop sensor (photodiode and LED) is also installed in the regulator area to detect the number of drops in real time. In the middle of the pole there is a 17 cm x 15 cm System Box, which functions as the system control center. Inside this box are installed electronic components such as ESP32, communication module, LCD, buzzer, and keypad. The LCD will display information on volume and drip-rate, while the keypad allows users to enter settings manually. To ensure stability during use, the infusion pole is designed with a 50 cm x 4 cm cross-shaped support base, which is equipped with wheels for mobility. This overall physical design not only supports the technical performance of the infusion automation system, but also considers ergonomic and safety aspects in use in medical environments such as hospitals or clinics.

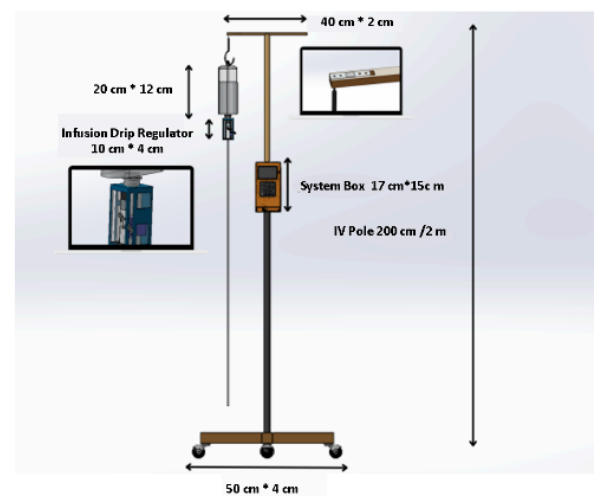


Figure 3. Physical Design of Infusion Device

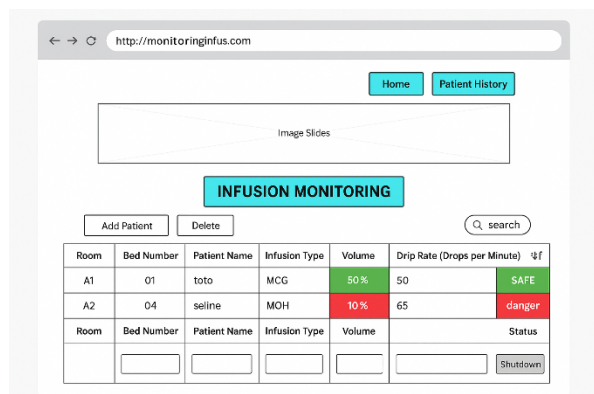


Figure 4. Software Design: Dashboard Monitoring

The monitoring dashboard, illustrated in Figure 4, is a web-based interface developed to enable real-time monitoring of patient infusions. There are navigation menus such as Home and Patient History, as well as a video display for visualization of the device's condition. Users can easily add, delete, and search for patient data. Patient data is displayed in a table that includes information such as room, patient name, infusion type, infusion volume, drip-rate, and status. The infusion volume is displayed as a percentage with different

colors as a level indicator, while the status indicates whether the infusion condition is normal or critical. This system is integrated with IoT devices via MQTT and a database, so that information is always updated automatically and facilitates quick decision-making by medical personnel.

2.6. Prototype Development

After the design stage is complete, the next step is to create a prototype. The goal of this stage is to realize the system design into a real device that can work according to its function. The prototype is assembled by arranging and connecting electronic components on a breadboard or PCB, so that the system can be tested and calibrated before it is actually used permanently. This prototype combines various sensors, actuators, microcontrollers, and communication modules to create a smart and responsive infusion system. The prototype of the IoT-based automatic infusion monitoring and control system developed in this study is shown in Figure 5. The system integrates key components to monitor and regulate the infusion drip-rate in real-time. A Load Cell, mounted at the top of the infusion pole, accurately measures the weight of the infusion fluid to determine the remaining volume. The Infusion Tube and Drip Chamber serve as the fluid flow path and the site for drop detection. The Brain System, consisting of an ESP32 microcontroller and sensors, processes inputs from the photodiode (for drip-rate) and load cell (for fluid weight). Monitoring results are displayed in real-time via a computer-based Dashboard. All electronics are housed in the Case Main Controller, containing two ESP32 units, an HX711 module, a servo motor control circuit, and communication modules. The servo motor adjusts the drip-rate automatically based on sensor data.

Figure 5. illustrates a well-integrated and functional system, designed to deliver accurate and reliable infusion control suitable for medical applications.

Its main functions include: **Measuring Infusion Fluid Volume:** With the help of the Load Cell sensor and the HX711 module, the system can measure the weight of the remaining fluid in the infusion bag. This weight value is then converted into volume, which is used to determine the condition of the infusion contents. **Detecting and Displaying Drip Rate:** Photodiode and LED sensors are used to count the number of drops that pass through. The results are displayed in drops per minute on the LCD screen. **Adjusting Drops Automatically:** The system will adjust the drip rate using a servo motor controlled by the ESP32. This motor opens or closes the infusion valve so that the drip rate remains in accordance with the medical target. **Sending Data to Server Online:** All data, such as drip rate and fluid volume, are sent directly to the server via the MQTT protocol, then stored in a MySQL database. This allows medical personnel to monitor the condition of the infusion remotely via a web dashboard. **Notification via LCD, Buzzer, and Web:** The system is equipped with a buzzer and status display via a web interface to provide warnings if there is a problem, such

as no drips within a certain time, fluid is running out, or the drip-rate is not as specified.

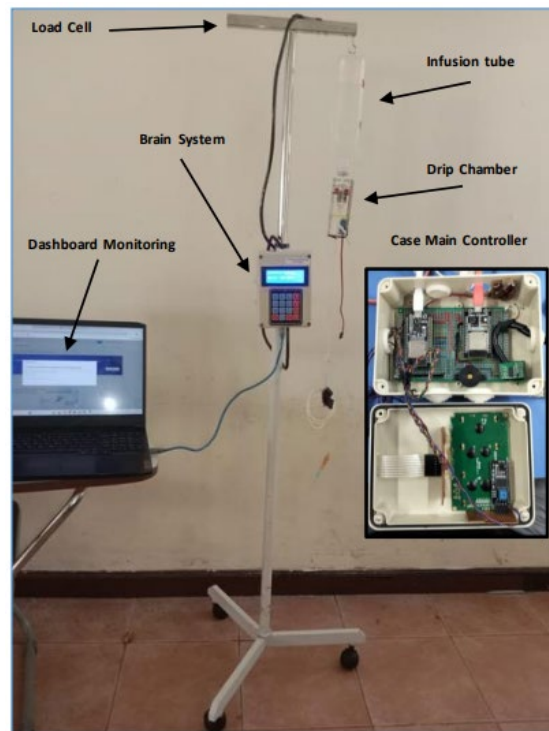


Figure 5. Development of an Infusion System Prototype

3. Results and Discussions

This section discusses the results of the prototype testing that has been developed, especially related to the accuracy of the Load Cell sensor and the analysis of data obtained during the test. The testing of the automatic infusion monitoring and control system was conducted in a campus laboratory environment, specifically in the Embedded System and IoT laboratory under the Computer Technology study program. The testing followed an experimental approach using hardware assembled to resemble real hospital conditions, including the use of an infusion bottle, photodiode, LED, load cell, and an ESP32 microcontroller connected to a Wi-Fi network. The simulation covered the monitoring of infusion drop rate per minute, detection of infusion fluid volume, and the system's response to abnormal conditions such as a halt in drip flow or a nearly empty infusion bottle. The system was tested to ensure that the buzzer alarm and LCD display functioned correctly according to the predefined parameters. Although the system has not yet been tested directly in a hospital setting, the design and simulation conditions were carefully configured to mimic real-world scenarios. Long-term testing has not yet been conducted to evaluate the system's durability under continuous use, nor has the impact of external factors such as temperature, humidity, or electromagnetic interference on the performance of the load cell sensor been assessed. Further testing in an actual medical environment is planned as a next step for clinical validation and system improvement.

3.1 Load Cell Sensor Accuracy

Accuracy testing on the Load Cell sensor is carried out to ensure that the sensor can read mass accurately and consistently. In this system, the sensor plays an important role in measuring the volume of infusion fluid based on weight, so that the accuracy of the reading is a factor that greatly determines the overall performance of the system.

The test was carried out using a standard load of 50 grams. The load was placed on the Load Cell 10 times, and each reading result was compared with the actual value. Furthermore, the percentage of error was calculated using Formula 5.

$$\% \text{ error} = \frac{\text{Sensor Result} - \text{Actual Weight}}{\text{Actual Weight}} \times 100\% \quad (5)$$

Based on the results of the accuracy measurements presented in Table 4, it can be concluded that the load

cell sensor demonstrated consistent readings, as all tests used the same load of 50 grams. The reading results from the sensor showed numbers that were very close to the actual value, ranging from 49.98 grams to 50.03 grams. This shows that the sensor has very good consistency in providing measurement results.

Table 4. Accuracy Test Results of the Load Cell Sensor

Testing	Load Weight (G)	Load Cell Sensor (G)	Error (%)
T-01	50	50.03	0.06
T-02	50	50.01	0.02
T-03	50	50.03	0.06
T-04	50	49.98	0.04
T-05	50	50.02	0.04
T-06	50	49.99	0.02
T-07	50	49.99	0.02
T-08	50	50.01	0.02
T-09	50	49.98	0.04
T-10	50	49.98	0.04

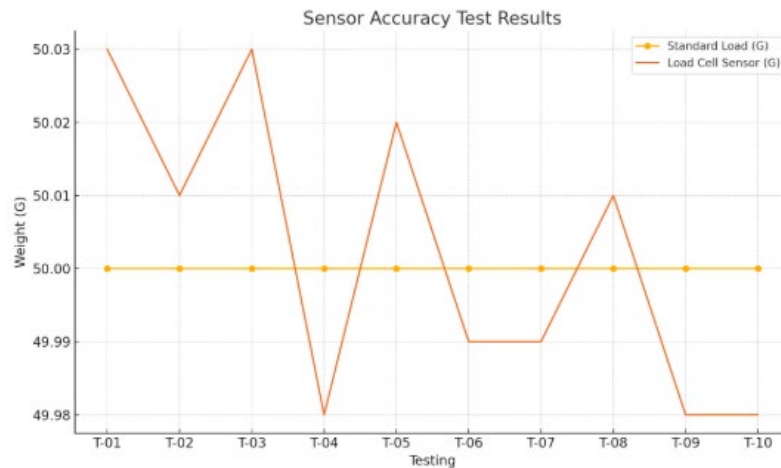


Figure 6. Sensor Accuracy Test Results

Figure 6 shows the error rate observed during testing, which ranged from 0.02% to 0.06%. This very low error rate indicates that the sensor has a high level of accuracy. There was no significant difference between the actual weight and the reading from the sensor. Performance Stability: Low and almost unchanged error indicates that the sensor has good stability when used repeatedly. There are no symptoms of fluctuation or interference in the measurement results.

Conclusion: Based on the tests that have been carried out, it can be concluded that the Load Cell sensor used in this system has: High accuracy in reading the load, Good consistency of measurement results (reproducibility), A very low error rate and can be ignored for practical purposes.

3.2 Testing of Infusion Volume Detection and Monitoring System

Real-time monitoring of infusion fluid volume is an important part of ensuring that patients receive the right amount of fluid as needed. By utilizing the ESP32 microcontroller and load cell sensor, changes in weight in the infusion tube—which directly correlates to the

volume of fluid inside—can be detected and monitored continuously.

This test aims to evaluate the accuracy of the system in detecting and monitoring the remaining infusion fluid volume. The load cell sensor connected to the ESP32 is used to read weight data in real time, which is then converted into volume information and displayed via LCD or Serial Monitor. The test step is carried out by filling the infusion tube using a previously known volume of fluid (for example 500 mL).

The system is activated to monitor the decrease in volume along with the infusion flow, and the readings from the system are compared with manual measurements. The main goal is to ensure that the system is able to provide accurate information, as well as react quickly to changes in volume. From observations, the system has proven to have a good response time, with data updates every one second—fast enough for medical needs.

As shown in Figure 7, the system successfully identifies and classifies the weight and volume status of infusion fluids in real time and displays the information clearly

on the LCD, making it easily accessible to both medical personnel and patients.

This implementation is important in improving patient safety and efficiency in terms of infusion management in a hospital or clinic environment.

As illustrated in Figure 8, the scatter plot of 200 volume test samples is visualized using color-coded notification levels: green for FULL (>350 mL), orange for HALF (150–350 mL), and red for WARNING (<150 mL). In conducting the Volume Detection and Volume Monitoring tests, the author created 200 volume test samples. The results show consistency, and data from 10 of these samples are presented in the Table 5.

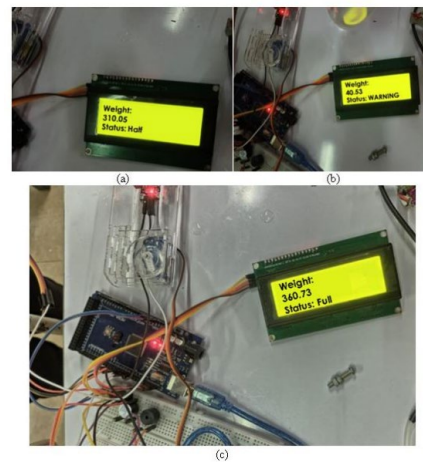


Figure 7. Monitoring of Infusion Volume and Status

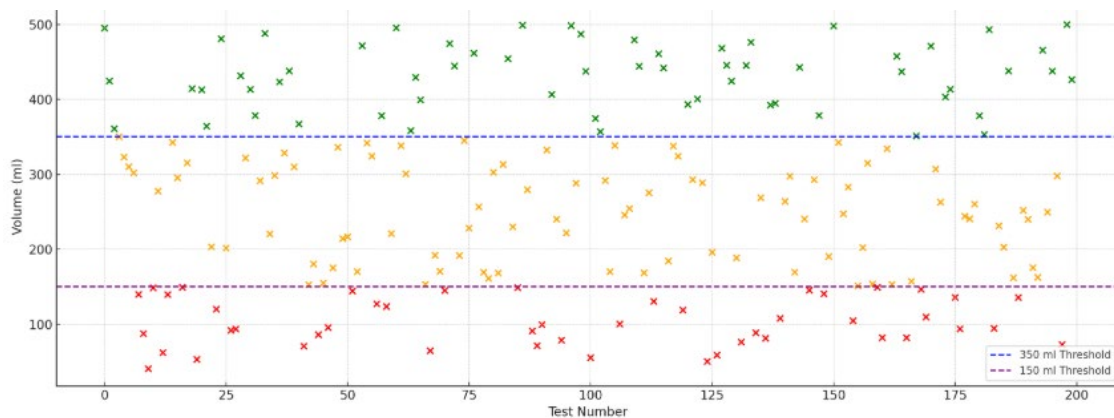


Figure 8. Scatter plot visualizes all 200 volumes test samples

Table 5. Results of the Infusion Volume Detection and Monitoring Test

Testing	Volume (ml)	Status (LCD)	Notification (Buzzer)
Test-01	495.05	FULL	Buzzer OFF
Test-02	424.41	FULL	Buzzer OFF
Test-03	360.73	FULL	Buzzer OFF
Test-04	350.00	HALF	Buzzer OFF
Test-05	322.90	HALF	Buzzer OFF
Test-06	310.05	HALF	Buzzer OFF
Test-07	301.74	HALF	Buzzer OFF
Test-08	139.67	WARNING	Buzzer ON
Test-09	87.34	WARNING	Buzzer ON
Test-10	40.53	WARNING	Buzzer ON

As shown in Table 5, the system demonstrated consistent performance in line with infusion volume classification standards. In Test-01 to Test-03, the infusion volume was identified as greater than 350 mL, corresponding to the FULL status. The system displayed the status "FULL" on the LCD, and the buzzer was inactive—indicating a normal condition because the fluid was still sufficient. Test-04 to Test-07: The infusion volume was detected to be between 150 mL and 350 mL. The system gave the status "HALF", and the buzzer remained silent. This indicates that although the fluid was reduced, no replacement action was needed. Test-08 to Test-10: The detected volume was below 150 mL. The system automatically displayed the status "WARNING" and the buzzer was activated as an alarm, indicating that the infusion fluid was running

low and needed to be replaced immediately. The test results showed that the infusion volume monitoring system worked with a high level of accuracy and responsiveness. The LCD indicator and notification via buzzer functioned according to the predetermined volume parameters. No significant differences were found between the sensor readings and the actual volume conditions, indicating that this system is reliable enough to support the automatic infusion monitoring process and help improve patient safety in administering intravenous fluids.

3.3 Testing of Automatic Control of Infusion Fluid Drip-Rate

Testing of automatic control of infusion fluid drip-rate is a critical process to ensure that the infusion control system works properly and accurately. The purpose of this test is: Validating the system's ability to regulate the infusion fluid drip-rate according to the expected value. Observing the system's response to variations in flow conditions, either too fast or too slow. The testing methodology involves evaluating the system at various drip-rate set points (in drops per minute). A photodiode is used as a sensor to measure the actual drip rate, which is then compared with the tolerance range ($\pm 5\%$) of each set point. Based on the measurement results, the system automatically adjusts the servo motor angle to regulate the fluid flow.

The tolerance range values in Table 6 were determined using Equations (2), (3), and (4). Equation (2) calculates the upper limit (Rupper), Equation (3) defines the lower limit (Rlower), and Equation (4) specifies the overall tolerance range from Rlower to Rupper. These equations provide the reference boundaries for evaluating the drip rate accuracy. The results of the automatic control test for the infusion fluid drip rate are shown in Table 6.

Figure 9 illustrates the comparison between the target drip rate and the values measured by the photodiode sensor for each test scenario, including action annotations (notifications) such as Increase, Decrease, and Normal Servo Angle. The notification color indicates the servo action: Green for Normal, Red for Decrease, Blue for Increase. As shown in Table 6, the automatic system demonstrated its ability to regulate the infusion drip rate using a servo motor and photodiode sensor. The testing process is carried out based on control logic. The test was carried out 100 times and then the author presents data from 10 tests in this case. Drip-Rate (Set Point) is the target value of the infusion drip-rate that has been previously determined (in drops per minute). This value is the main reference in testing. Each drip-rate is given a certain tolerance range, namely the upper and lower limits of the drip-rate value that is still considered acceptable. This range allows for slight variations in sensor readings, without being considered an error. Photodiode Sensor Value is the actual value of the drip-rate read by the photodiode

sensor. This value is then compared to the tolerance range that has been set. Based on the results of the sensor reading, the system provides a notification in the form of an automatic control action on the servo motor angle. The Notification column in the table shows the corrective action or automatic system response to the results of the photodiode sensor reading compared to the drip-rate tolerance range. This notification functions as an indicator of the control decision on the servo motor angle and aims to keep the infusion drip-rate stable according to the target.

Table 6. Test Results of Automatic Control of Infusion Drip-Rate

Testing	Drip-Rate	Range Tolerance	Photodiode Sensor Value	Notification
Test-01	62.5	59.38 - 65.62	62.0	Normal Servo Angle
Test-02	15.6	14.82 - 16.38	17.0	Increase Servo Angle
Test-03	20.8	19.76 - 21.84	18.5	Decrease Servo Angle
Test-04	26.6	25.27 - 27.93	26.4	Normal Servo Angle
Test-05	41.6	39.52 - 43.68	41.0	Normal Servo Angle
Test-06	42	39.90 - 44.10	45.8	Increase Servo Angle
Test-07	55.6	52.82 - 58.38	52.3	Decrease Servo Angle
Test-08	33.3	31.63 - 34.96	30.0	Decrease Servo Angle
Test-09	44.4	42.18 - 46.62	47.0	Increase Servo Angle
Test-10	62.5	59.38 - 65.62	66.0	Increase Servo Angle

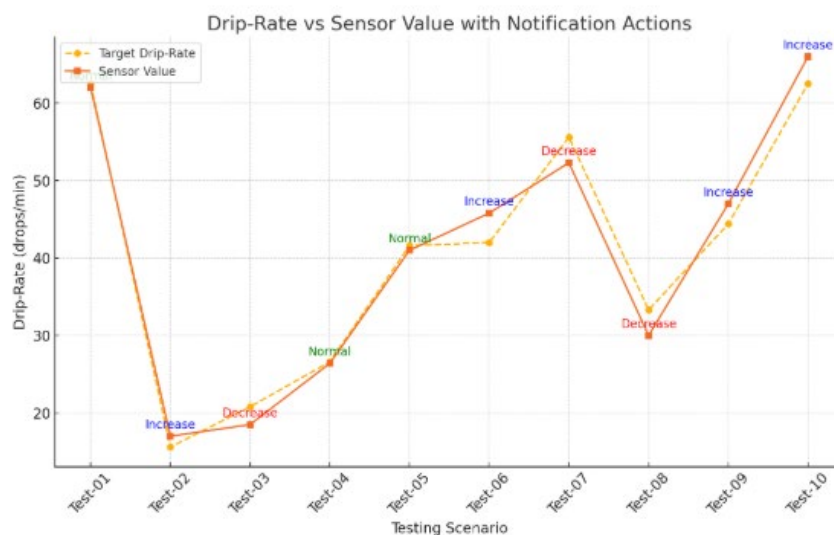


Figure 9. Drip-Rate vs Sensor Value with Notification Actions

Analysis of Automatic Control Testing of Infusion Fluid Drop Rate: Normal Servo Angle Condition, When the sensor reading shows a value that is still within the specified tolerance limit, the system interprets it as a condition where the infusion fluid drip-rate is in accordance with the target. In this condition, the system does not make changes to the servo angle because the fluid flow is considered stable and according to needs. For example, in Test-01 with a target of 62.5

drops/minute and a sensor reading of 62.0 drops/minute, the difference is still within reasonable limits. The same thing also happened in Test-04 and Test-05, where the difference between the target and the detection results was still within the permissible range. This shows that the system can maintain the stability of the infusion rate without the need for further adjustments.

Servo Angle Addition (Flow Too Fast): If the value read by the sensor exceeds the upper tolerance limit, the system will detect that the drip-rate is too fast. To slow the flow, the system will slowly increase the servo angle, which means opening the valve less so that the fluid rate decreases. For example, in Test-02, the drop rate target is 15.6 drops/minute, but the sensor reads 17.0 drops/minute. The same thing happened in Test-06, Test-09, and Test-10, where all showed a flow rate that was too high. In all of these tests, the system was able to respond appropriately by reducing the hose opening, so that the flow returned to the specified parameters.

Servo Angle Reduction (Flow Too Slow): Conversely, if the sensor detects a drop rate below the tolerance threshold, the system considers the fluid flow to be too slow. To overcome this, the servo angle is reduced so that the infusion valve opening becomes wider and the fluid can flow faster. Based on testing, this happened in Test-03, Test-07, and Test-08. In all cases, the system was able to recognize that the flow was slower than it should be, then took action by opening the hose larger until the drop rate returned to the specified target.

From all test results, this automatic control system has proven to be able to work consistently and accurately in managing the drip-rate of infusion fluid. When the rate is too fast, the system automatically reduces the opening (increases the servo angle) to reduce the flow rate. When the rate is too slow, the system opens the hose wider (decreases the servo angle) to speed up the flow. When the rate is within the tolerance limit, no adjustments are made because the condition is stable.

With the right response to each condition, this system is very reliable in automatically maintaining the stability of the infusion rate according to the predetermined medical parameters.

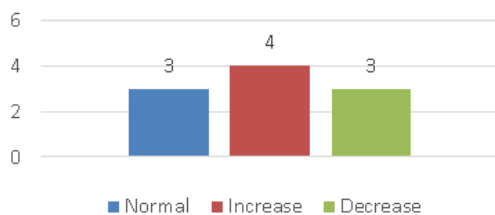


Figure 10. Testing of Automatic Control of Intravenous Fluid Drip-Rate

Figure 10 presents a bar chart of the test results of the automatic infusion drip rate control system, categorized into three types of system actions in response to sensor readings. (1) Normal (blue): The system detected 3 times that the drip rate was in accordance with the target (within the tolerance range), so there was no need for adjustments to the servo angle. (2) Increase (red): The system detected 4 times that the drip-rate was too fast, so the system responded by increasing the servo angle to slow down the flow of infusion fluid. (3) Decrease (green): The system read 3 times that the drip-rate was too slow, so the servo angle was reduced to enlarge the opening and speed up the infusion flow.

In conclusion, this graph shows that the system can dynamically and responsively adjust the infusion drip-rate according to the conditions detected by the sensor. The “Increase” category is the most frequently performed action, indicating a tendency for infusion drips to be too fast and require correction.

In the future, it is suggested that this system be enhanced by incorporating artificial intelligence (AI) [24] - [28]. By leveraging AI, the system would be capable of analyzing past usage patterns to automatically regulate the drip rate and more accurately forecast fluid requirements.

3.4 Limitations and Future Work

As a reflection on the development of the system, there are several potential limitations that need to be considered in implementing this system in a real clinical environment: **Calibration Challenges:** The accuracy of the servo motor in regulating the drip-rate depends heavily on proper calibration with different types of infusion sets. While initial calibration was successful under controlled conditions, variability in the physical setup (e.g., fluid viscosity, tubing diameter, or valve stiffness) may require recalibration for each clinical use case. **Sensor Sensitivity and Environmental Factors:** Photodiode sensors used to detect drip-rate are susceptible to interference from ambient light and physical disturbances, which could affect reliability in busy clinical environments. Similarly, the load cell’s accuracy may degrade slightly if not properly mounted or if subjected to strong vibrations. **System Limitations in Real-World Settings:** Although the prototype performed well in simulated scenarios, its full reliability and durability in long-term clinical use have not yet been evaluated. **Integration with hospital systems (e.g., electronic medical records)** and hygiene compliance (e.g., waterproofing, sterilization) also require further study. **System Integration and Interoperability:** The system has not yet been fully integrated with existing hospital information systems, such as Electronic Medical Records (EMR) or digital infusion management systems. Software adaptation and compatibility testing are required to ensure that the system can operate harmoniously without disrupting existing medical workflows. **Technical Challenges in the Field:** In field implementation, several technical challenges may arise, including the stability of Wi-Fi networks in hospital areas, the need for backup power supply (e.g., UPS) to prevent disruptions caused by power outages, and potential electromagnetic interference from other medical equipment. Additionally, training medical staff to use the system optimally is also an important factor that has not been fully addressed.

In the future, the development of this system is recommended to integrate artificial intelligence (AI): With AI support, the system can learn from previous usage patterns to automatically adjust the drip-rate and predict fluid needs more precisely. AI also plays a role in analyzing the large amount of data collected by the

system, so that it is able to provide data-based medical recommendations. This development will improve the performance and flexibility of the system, as well as enable more sophisticated and efficient remote infusion monitoring, especially in a hospital environment. Miniaturization and Enclosure Design: For real-world hospital use, hardware enclosure refinement is needed to ensure portability, safety, and ergonomic usability. Extended Clinical Testing: Future work will include testing in actual hospital environments, with medical personnel feedback to validate usability and reliability. Battery Backup and Fail-safe Mechanisms: To prepare for power outages or connectivity loss, backup power and local alert systems will be considered. ANOVA or regression analysis for drip-rate stability under different servo angles and patient conditions.

4. Conclusions

This research successfully designed and built a system that combines automatic infusion monitoring and control functions in one Internet of Things (IoT)-based device. The system utilizes various main components, such as the ESP32 microcontroller, load cell sensor, photodiode sensor, and servo motor, which work in a coordinated manner to monitor the volume of infusion fluid while automatically regulating the drip-rate. From the test results, the load cell sensor showed a very high level of accuracy with a very small margin of error, which is between 0.02% and 0.06%. This ensures that the measurement of the infusion fluid volume reflects the actual condition. In addition, the system is able to display the fluid status in real-time via LCD and Web Monitoring, with the categories "FULL", "HALF", or "WARNING", and provide a warning using a buzzer if the infusion volume is running out. In terms of automatic control, the system is able to respond quickly to changes in the drip rate, either by adjusting the angle of the servo motor to slow down or speed up the flow of fluid, so that the stability of the infusion rate is maintained according to the predetermined target value. The test results show that this system is able to work optimally in maintaining the stability of the infusion fluid flow, helping to improve patient safety, and reducing the risk of human error in the infusion setting process. With the application of IoT technology, this system is very relevant for use in treating patients with certain medical conditions, such as those with acid reflux or typhoid, who require consistent and accurate infusion monitoring. This research makes a significant contribution by presenting a more efficient, safer, and more responsive solution for patient care in response to medical needs. From a literature perspective, the developed system provides a solid foundation for future research in the field of smart infusion technology based on the Internet of Things (IoT). The integrated design which combines real-time monitoring, automatic drip-rate control, and network-based notification features serves as an innovative reference for the development of similar systems in the future.

In terms of practical application, this system shows

great potential for implementation, particularly in healthcare facilities with limited resources. However, it must be acknowledged that the system is still at the prototype stage and not yet fully comparable to advanced systems already implemented in developed countries, especially in terms of stability, clinical validation, and scalability. Therefore, this study also emphasizes the importance of further development to enhance the system's features so that it can deliver more immediate and significant impact in supporting the work of medical personnel in real-world settings.

With its adaptive approach and design tailored to local needs, this system creates opportunities for the digital transformation of infusion services in hospitals, while also encouraging collaboration between researchers, medical professionals, and technology developers to produce effective and practical solutions.

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